Mixed-Frequency Approaches

Why Mixed-Frequency Models?

- Forecast targets often observed at low frequency (e.g., quarterly GDP, quarterly inflation).
- But many predictors are available at higher frequency (monthly surveys, daily financial data).
- Problem: Standard time-series models require matching frequencies $\Rightarrow$ throw away information or aggregate predictors.
- Solution: Mixed-frequency models use high-frequency indicators *directly*, without discarding data.
- Two main approaches:
  - State-space: treat missing low-frequency values as latent, handle via Kalman filter.
  - MIDAS regressions: regress directly on high-frequency lags with parsimonious lag polynomials.

MIDAS Regression: Basic Form

- Schematic regression:

$$y_t = \beta_0 + \sum_{k=0}^{K} \beta(k;\theta)\, x_{t-k/m} + \varepsilon_t$$

  - y_t: low-frequency target (e.g., quarterly GDP).
  - $x_{t-k/m}$: high-frequency regressor (e.g., monthly PMI, $m = 3$ months per quarter).
  - $\beta(k;\theta)$: lag weights governed by a few parameters (e.g., exponential Almon).
- Avoids estimating a separate β for each high-frequency lag $\Rightarrow$ parsimonious and stable.
- Estimation: nonlinear least squares or Bayesian methods.
- Widely used for nowcasting: combining daily/monthly indicators with quarterly targets in real time.

References: Ghysels, Santa-Clara, Valkanov (2007); Ghysels et al. (2016). R package: `midasr`.

State-Space Mixed-Frequency Models

- Alternative: embed the mixed-frequency structure into a state-space model.
- Intuition:
  - Low-frequency series treated as observed only at some dates.
  - Missing values in between are handled as latent states.
  - Kalman filter/smoother integrates information from all available high-frequency predictors.
- Flexibility: can handle multiple predictors, dynamic interactions, and structural restrictions.
- Applications: central banks often use state-space mixed-frequency VARs for forecasting GDP and inflation.

Reference: Mariano & Murasawa (2003); Schorfheide & Song (2015).

Intuition: Quarterly GDP vs. Monthly Indicators

- Quarterly GDP is sparse: only one observation every three months.
- Monthly surveys/indicators provide more frequent signals.
- Mixed-frequency methods combine them:
- $\rightarrow$ “fill in the gaps” between quarterly data using higher-frequency information.

Mixed-Frequency VARs: Intuition

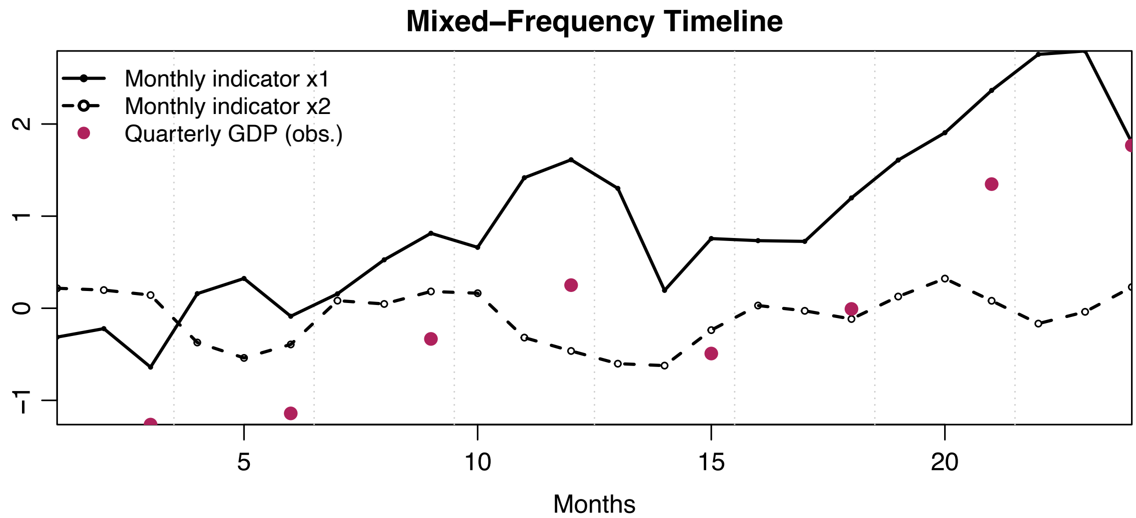
- Generalization of VAR models to allow variables sampled at different frequencies.
- Example: quarterly GDP, monthly inflation, daily financial indicators in one system.
- State-space form:
  - Observation equation handles missing values for low-frequency series.
  - Transition equation is the usual VAR dynamics.
- Estimation typically Bayesian (Minnesota priors, Gibbs sampling).

Reference: Schorfheide & Song (2015, Journal of Business and Economic Statistics).

Why MF-VARs?

- Captures interactions between multiple variables at mixed frequencies.
- Consistent forecasting framework:
  - Use monthly and quarterly data without aggregation.
  - Generate density forecasts for quarterly targets.
- Natural extension of the widely used VAR framework.
- Flexible: can incorporate priors, structural restrictions, or time variation.

MF-VAR Intuition: Timeline



- Quarterly GDP “fills in” only once per quarter.
- Monthly variables provide extra observations in between.
- State-space representation reconciles them within the VAR structure.